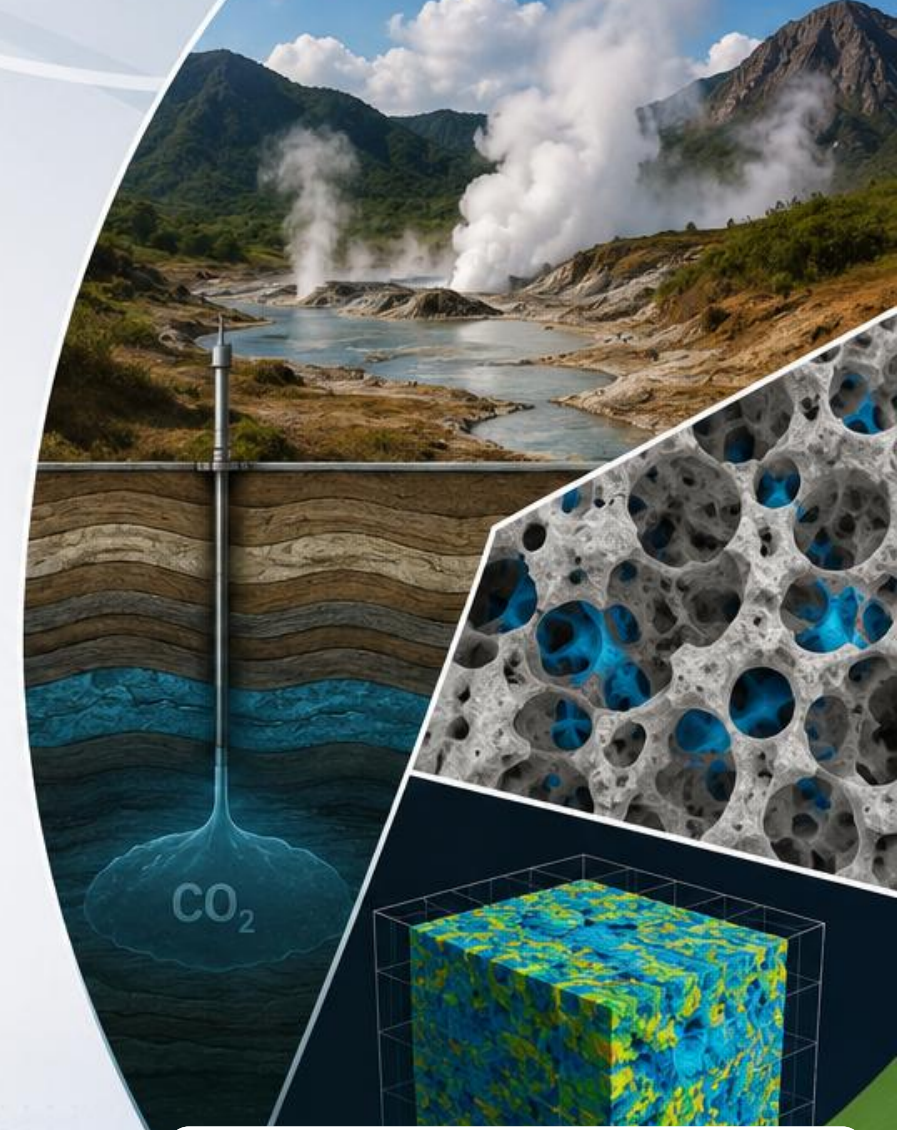
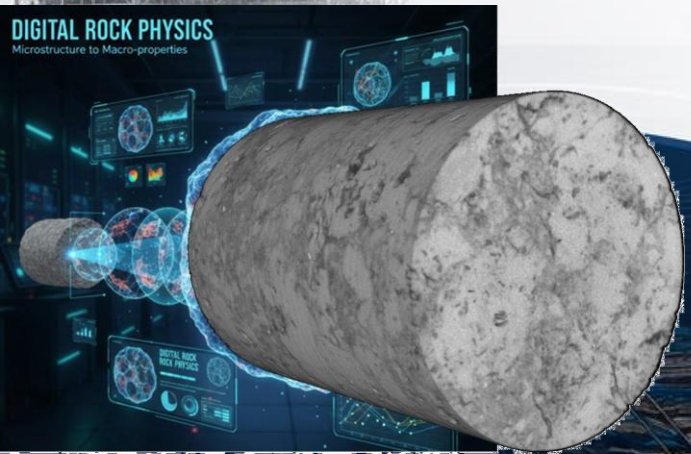




Digital Rock Physics

Workflow, Application &
Instrument Development



PHYSICS OF
POROUS MATERIALS
AND FLUID DYNAMICS
LABORATORY
PPMFD FMIPA - ITB

Imaging: Processing & Analysis

STRUKTURIN: workflow and in-house app for porous media analysis

Layer Properties

Volume Bounds (Cropping)

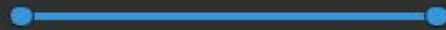
X-Axis (Width): 0 - 192



Y-Axis (Height): 0 - 299

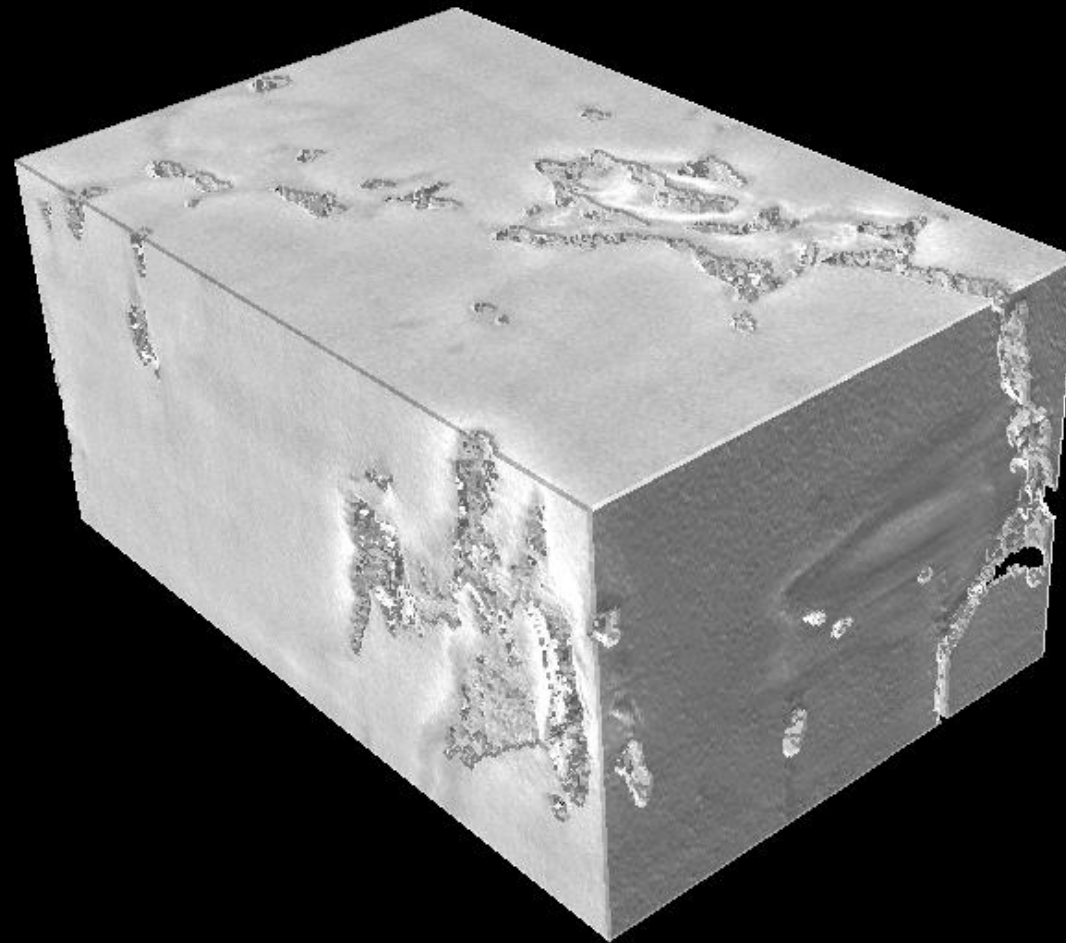


Z-Axis (Depth): 0 - 148



Show Histogram

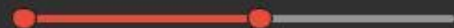
Material Settings

☒ Raw Grayscale Data

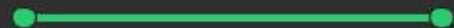
Layer Properties

Volume Bounds (Cropping)

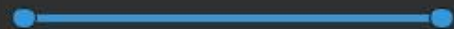
X-Axis (Width): 0 - 108



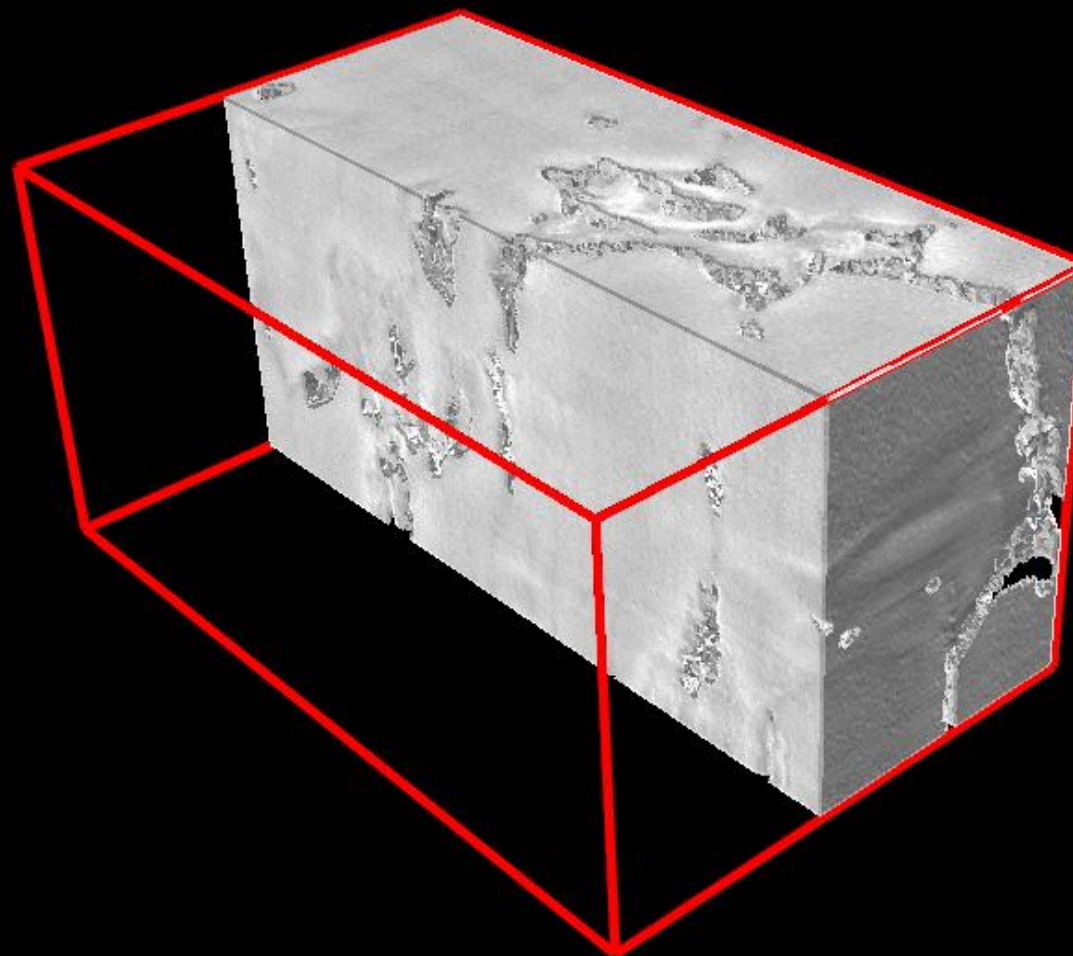
Y-Axis (Height): 0 - 299



Z-Axis (Depth): 0 - 148

 Show Histogram

Material Settings

☒ Raw Grayscale Data

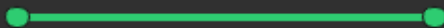
Layer Properties

Volume Bounds (Cropping)

X-Axis (Width): 0 - 96



Y-Axis (Height): 0 - 299

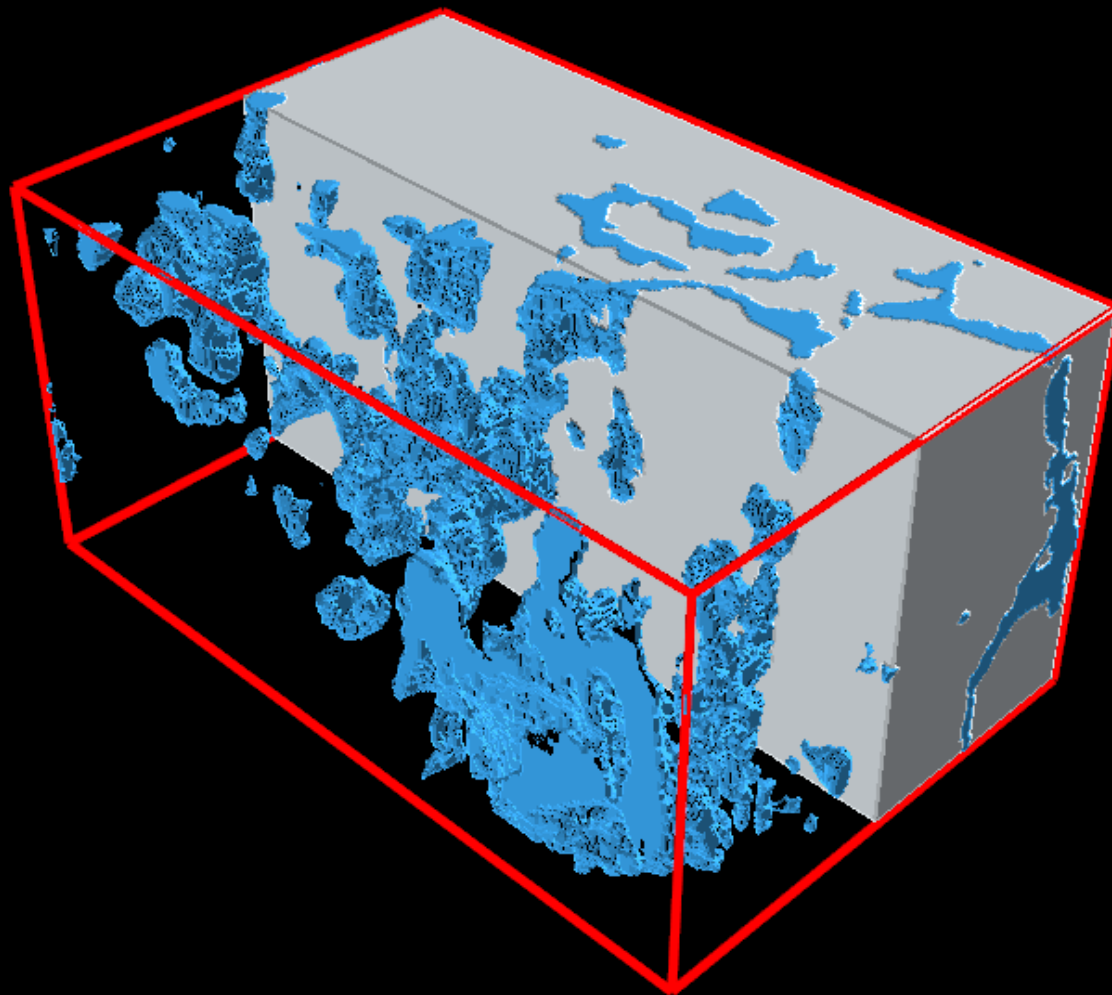


Z-Axis (Depth): 0 - 148



Show Histogram

Material Settings

☒ Pores (Mat 0)☒ Material 1PPMFD
Laboratory
FMIPA - ITB

Layer Properties

Volume Bounds (Cropping)

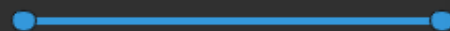
X-Axis (Width): 0 - 192



Y-Axis (Height): 0 - 299



Z-Axis (Depth): 0 - 148



Show Histogram

Material Settings

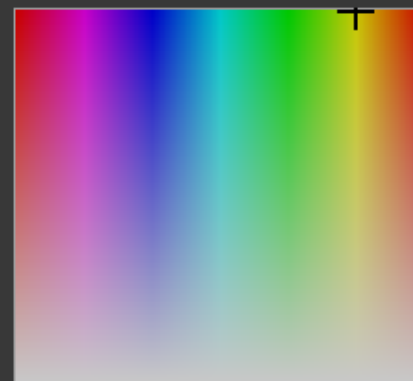
☐ Unmapped Pores☒ Z-Axis Paths☒ Y-Axis Paths☒ X-Axis Paths☐ Solid Matrix

Select Material Color

Basic colors



Pick Screen Color



Custom colors



Add to Custom Colors



Hue: 60

Red: 255

Sat: 255

Green: 255

Val: 255

Blue: 0

HTML: #ffff00

OK

Cancel

Layer Properties

Volume Bounds (Cropping)

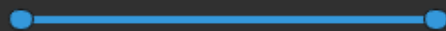
X-Axis (Width): 0 - 192



Y-Axis (Height): 0 - 299

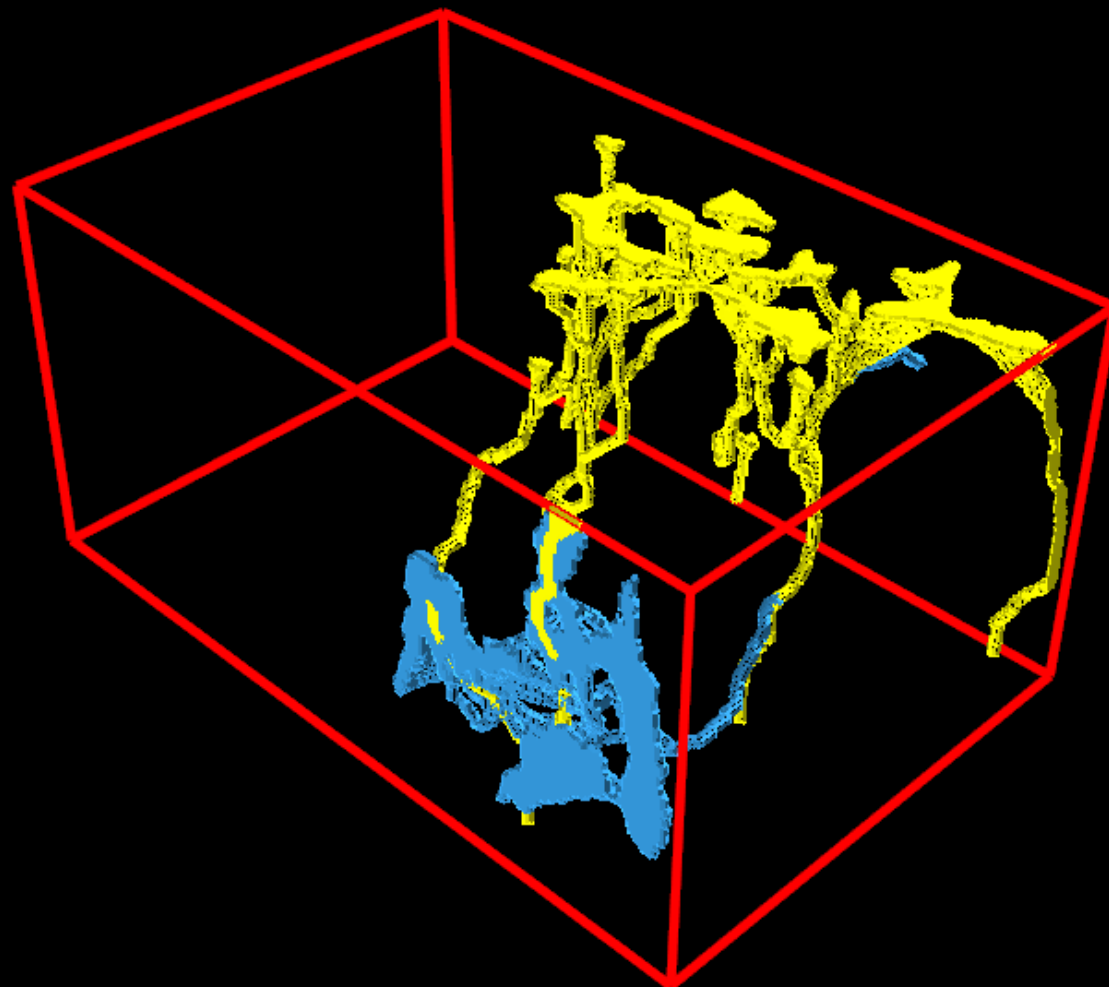


Z-Axis (Depth): 0 - 148



Show Histogram

Material Settings

☐ Unmapped Pores☒ Z-Axis Paths☒ Y-Axis Paths☒ X-Axis Paths☐ Solid MatrixPPMFD
Laboratory
FMIPA-ITB

Layer Properties

2D Tools

Slicer Display Crop ROI

X-Plane (YZ) 96



Y-Plane (XZ) 150

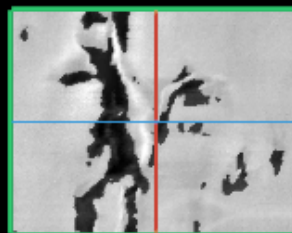
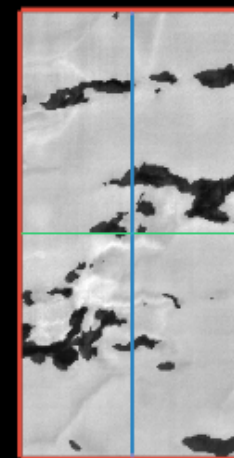
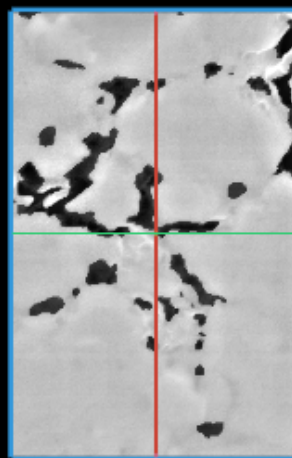


Z-Plane (XY) 74



Material Settings

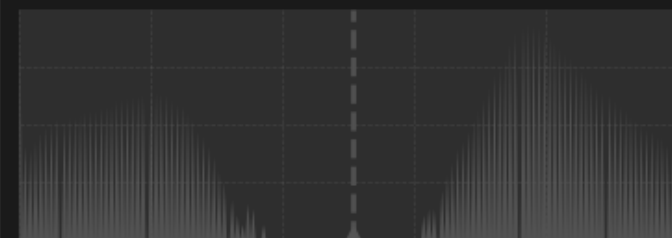
☒ Raw Grayscale Data



hresholdin ntrast Cur Color Map Metadata Line Profile

Cutoff: 127 Auto Preview Mask:

☐ Apply Global Threshold



Reset

Apply



PPMFD
Laboratory
FMIPA-ITB

Layer Properties

2D Tools

Slicer Display Crop ROI

X-Plane (YZ) 96



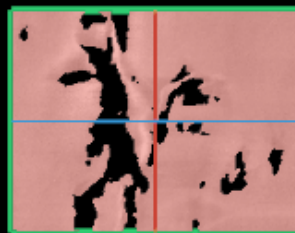
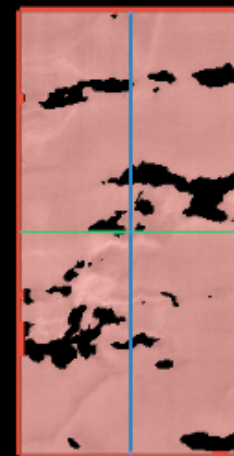
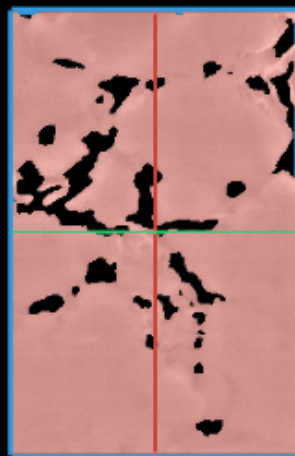
Y-Plane (XZ) 150



Z-Plane (XY) 74



Material Settings

☒ Raw Grayscale Data

hresholdin ntrast Cur Color Map Metadata Line Profile

Cutoff: 127 Auto Preview Mask: ☒ Apply Global Threshold

Reset

Apply

PPMFD
Laboratory
FMIPA-ITB

2D View

3D View

+ Zoom In

- Zoom Out

Reset Camera

Theme: Light

☒ Studio Lighting☐ Smooth Interpolation☒ Shadows:

Opacity:

Show Advanced Camera Tools ▾

Layer Properties

2D Tools

Slicer

Display

Crop

ROI

X-Plane (YZ)

96

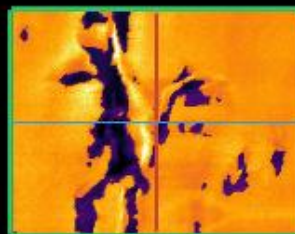
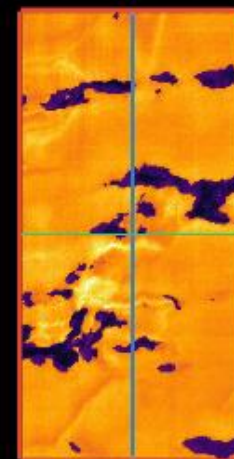
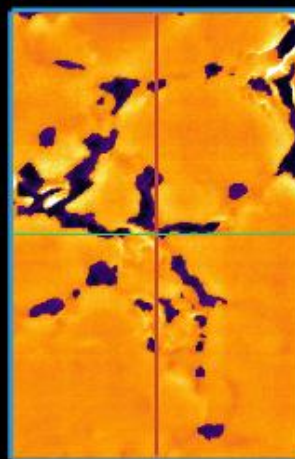
Y-Plane (XZ)

150

Z-Plane (XY)

74

Material Settings

☒ Raw Grayscale Datahresholdin ntrast Cur **Color Map** Metadata Line Profile

Min (Black): 0

Max (White): 255

LUT: Perceptual: Inferno

☐ Invert

- Original
- Grayscale
- Perceptual: Viridis
- Perceptual: Plasma
- Perceptual: Inferno
- Perceptual: Magma
- Standard: Jet
- Standard: Turbo
- Standard: Rainbow
- Heat: Hot (Black Body)

Reset

Apply



PPMFD
Laboratory
FMIPA-ITB

2D View

3D View

+ Zoom In

- Zoom Out

Reset Camera

Theme: Light

☒ Studio Lighting☐ Smooth Interpolation☒ Shadows:

Opacity:

Show Advanced Camera Tools ▾

Layer Properties

2D Tools

Slicer

Display

Crop

ROI

☒ Enable Cropping Box

Batas X:

X Size: 102 px (102.00 μm)

Batas Y:

Y Size: 300 px (300.00 μm)

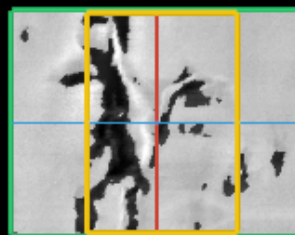
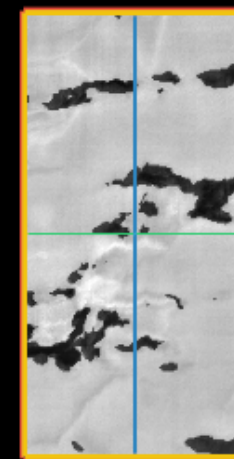
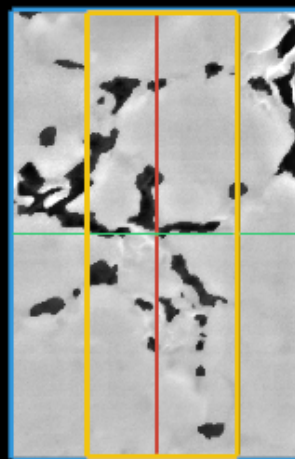
Batas Z:

Z Size: 149 px (149.00 μm)

Undo

Apply Crop

Material Settings

☒ Raw Grayscale Datahresholdin ntrast Cur Color Map Metadata Line Profile

Min (Black):

0

Max (White):

255

LUT: Original

☐ Invert

Reset

Apply

**PPMFD**
Laboratory
FMIPA - ITB

Layer Properties

2D Tools

Slicer

Display

Crop

ROI

☒ Enable ROI Masking

+ Add ROI

- Delete

Region 1

Color:



Cube (Box)

Arah Z

☐ Solid Fill (Transparent)☐ Invert (Keep Outside, Remove Inside)

Batas X:

129 px (129.00 μm)

Batas Y:

201 px (201.00 μm)

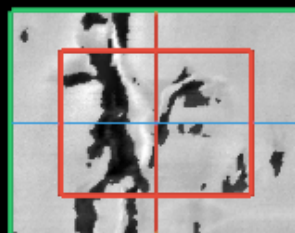
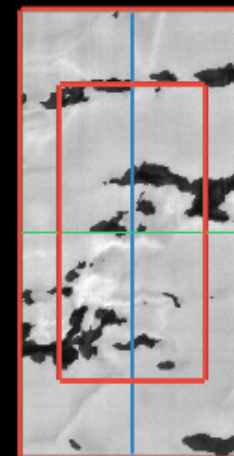
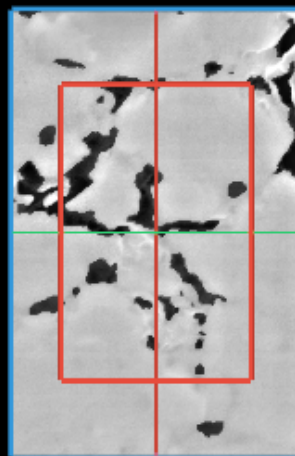
Batas Z:

99 px (99.00 μm)

Undo

Apply ROI

Material Settings

☒ Raw Grayscale Datahresholdin ntrast Cur Color Map Metadata Line Profile*Double Right-Click on the 2D plane to draw a line.***Physical Length:** 0.00 | **Pixel Distance:** 0 px**Resolution:** 0.00 | **Trajectory:** 0.0° (H), 0.0° (V)

Double Right-Click on the 2D plane to draw a line.

Clear Profile

**PPMFD**
Laboratory
FMIPA - ITB

Layer Properties

2D Tools

Slicer Display Crop ROI

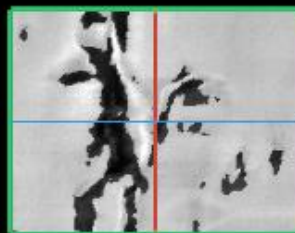
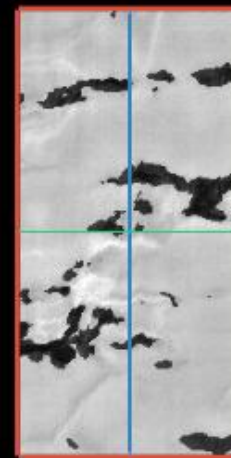
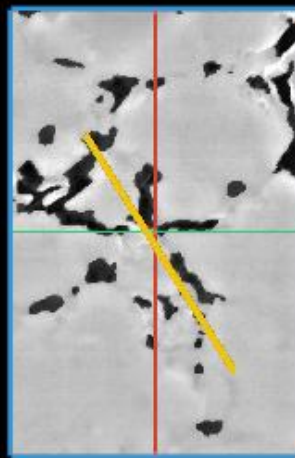
Viewport (1,1):

Histogram ▼

☐ Enable MPR Reorientation☒ Show Crosshairs Bounds☐ Show Orientation Axes 1,00☐ Show Plane Labels☒ Enable Line Profile Tool

Opacity:

Material Settings

☒ Raw Grayscale DataThreshold Contrast Cur Color Map Metadata Line Profile

Done. Double Right-Click to draw a new line.

Physical Length: 190.550 nm | Pixel Distance:

Resolution: 1 nm/px | Trajectory: 57.8° (H), 32.2



Clear Profile

Layer Properties

Volume Bounds (Cropped)

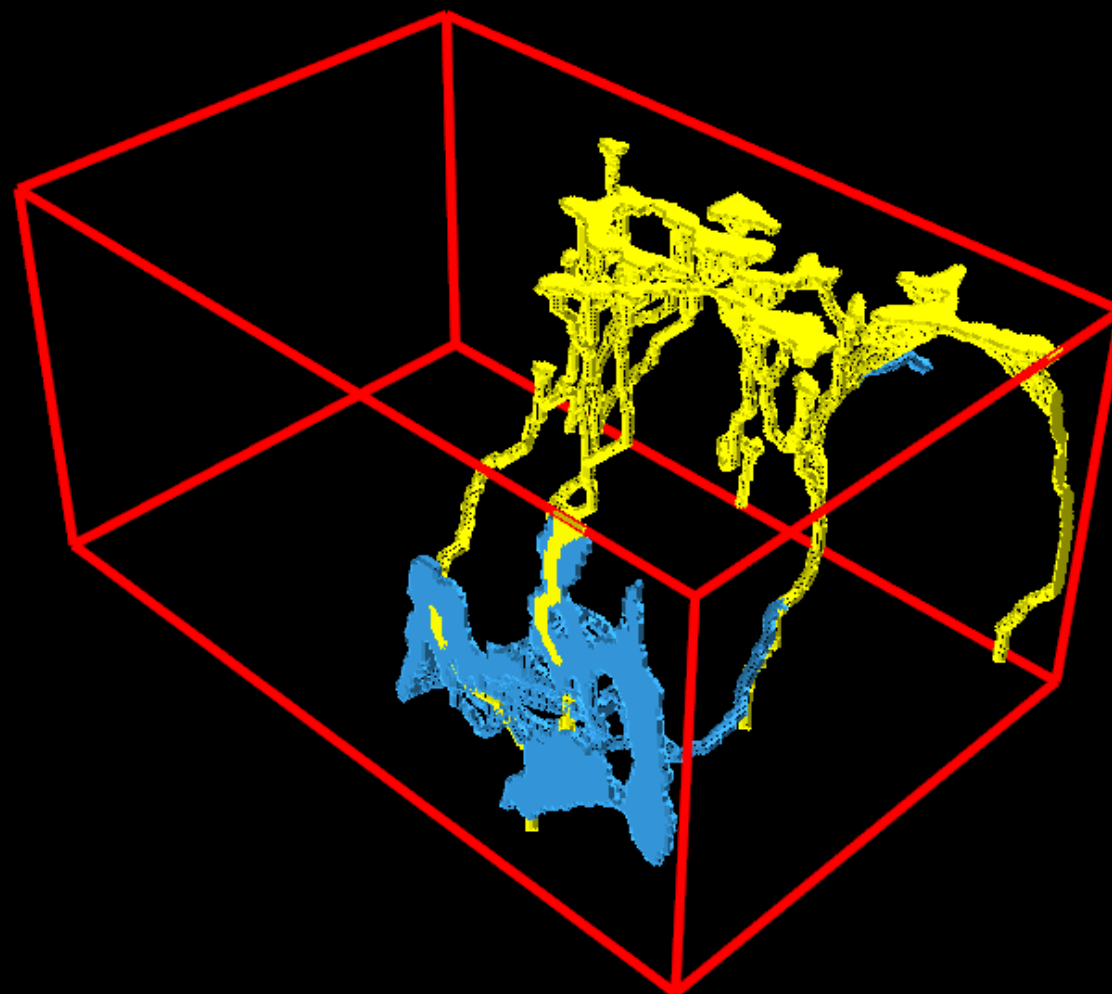
X-Axis (Width): 0 - 192

Y-Axis (Height): 0 - 299

Z-Axis (Depth): 0 - 148

Show Histogram

Material Settings

☐ Unmapped Pores☒ Z-Axis Paths☒ Y-Axis Paths☒ X-Axis Paths☐ Solid Matrix

PPMFD
Laboratory
FMIPA-ITB

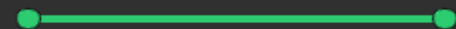
Layer Properties

Volume Bounds (Cropping)

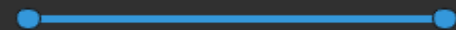
X-Axis (Width): 0 - 192



Y-Axis (Height): 0 - 299



Z-Axis (Depth): 0 - 148



Show Histogram

Material Settings

- ☐ Unmapped Pores
- ☒ Z-Axis Paths
- ☒ Y-Axis Paths
- ☒ X-Axis Paths
- ☐ Solid Matrix

DATA INPUT SETTINGS

Processing Mode: Manual (Single Folder/File)

Input Data Path: Manual (Single Folder/File)
Batch 1 (Parent -> Sub Folder)
Batch 2 (Parent -> Parent -> Sub Folder)

Output Folder: C:/Users/DigitalRockPhysics/Desktop/Summer School 2026/OR_7.5

Browse...

Output File Prefix: Analysis_Result

Volume Size: 149 slices (Z) x 300 (Y) x 193 (X)

Voxel Resolution: 1.000

mm/voxel

☐ Enable Dynamic Shrink-Wrap (Untuk Silinder/Ireguler)

RUN ANALYSIS

Ready to run...
Select mode: Manual or Batch, then click RUN.

Batch Analysis Summary Report

Total Samples: 6 | Generated on: 20 Aug 2026



Sample List

1	CM_7.5
2	DM_7.5
3	GF_7.5
4	GS_25
5	OR_7.5
6	SA_7.5

[Sampel List](#)

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Batch Analysis Summary Report

Total Samples: 6 | Generated on: 20 Aug 2026



Strukturin - Analysis Report

Data Source: CM_7.5

Resolution: 1.0 μm

Start Time: 20 Aug 2026 | 00:51:31

End Time: 20 Aug 2026 | 00:55:06

--- INTEGRATED ANALYSIS RESULTS ---

Total Porosity: 13.5410 %

Pore Surface Area: 2.0796e+06 μm^2

SSA (per Bulk Vol): 7.7024e-02 1/ μm

Z-Axis Results:

Through: 13.4428 %

Dead-end: 0.0318 %

Closed: 0.0664 %

Y-Axis Results:

Through: 13.4428 %

Dead-end: 0.0290 %

Closed: 0.0693 %

X-Axis Results:

Through: 13.4428 %

Dead-end: 0.0399 %

Closed: 0.0593 %

[Sample List](#)

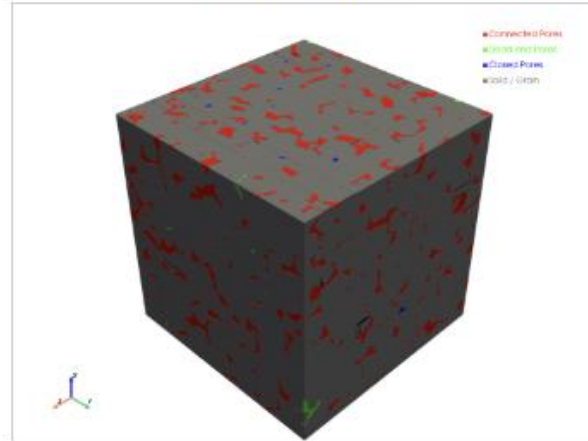
[< Previous](#)

Sample 1 of 6

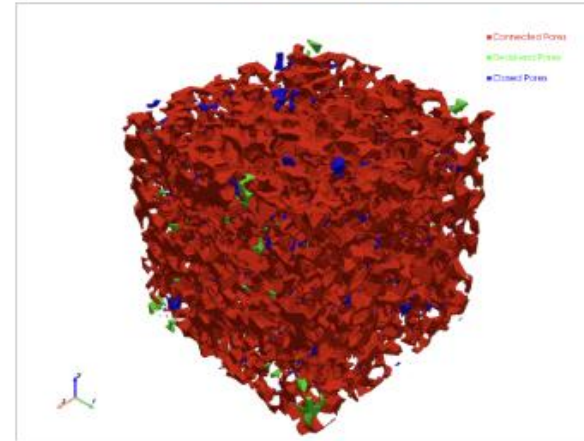
[Next >](#)

3D Visualization Summary

Full Bulk View



Pore Space View



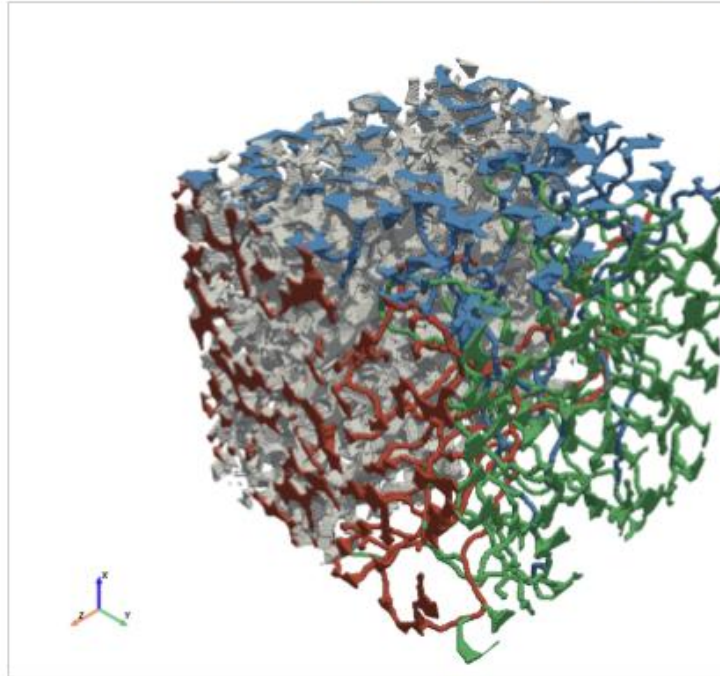
Through Pore [X]



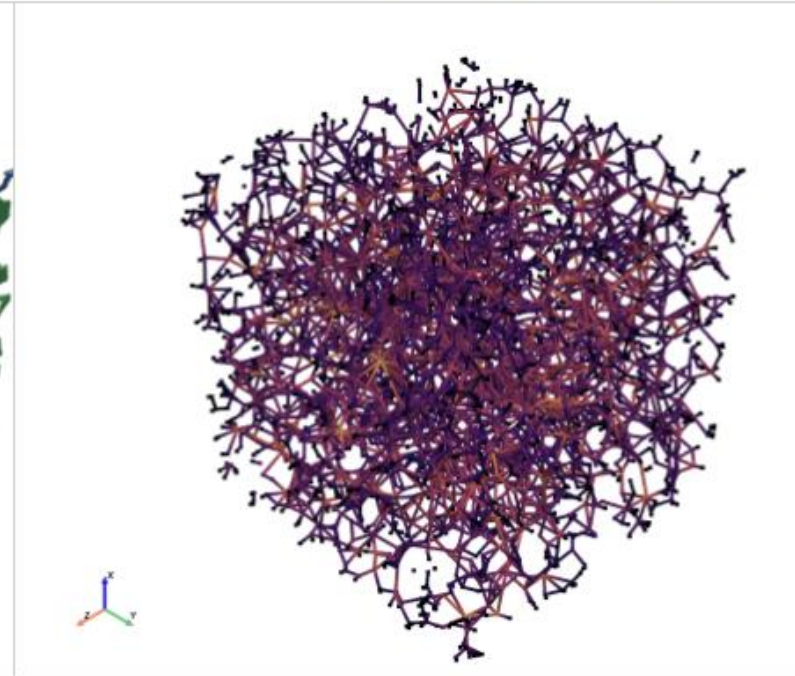
Through Pore [Y]



Cutaway View (Front Exposed)



PNM Flow Trajectory (Z-Axis)



Hardware System Logs:

Average Volumetric RAM Load : 8.53 GB

Average Active VRAM Load : 0.08 GB

[Sample List](#)

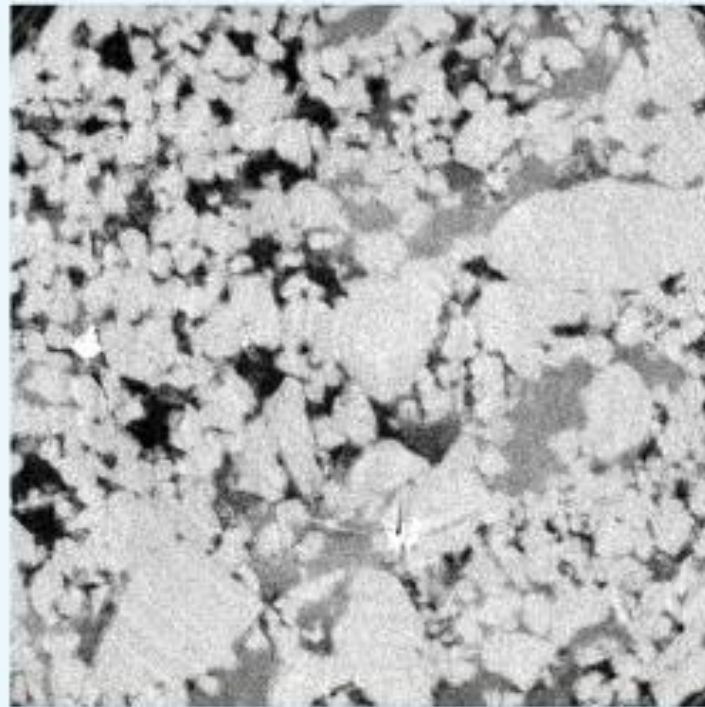
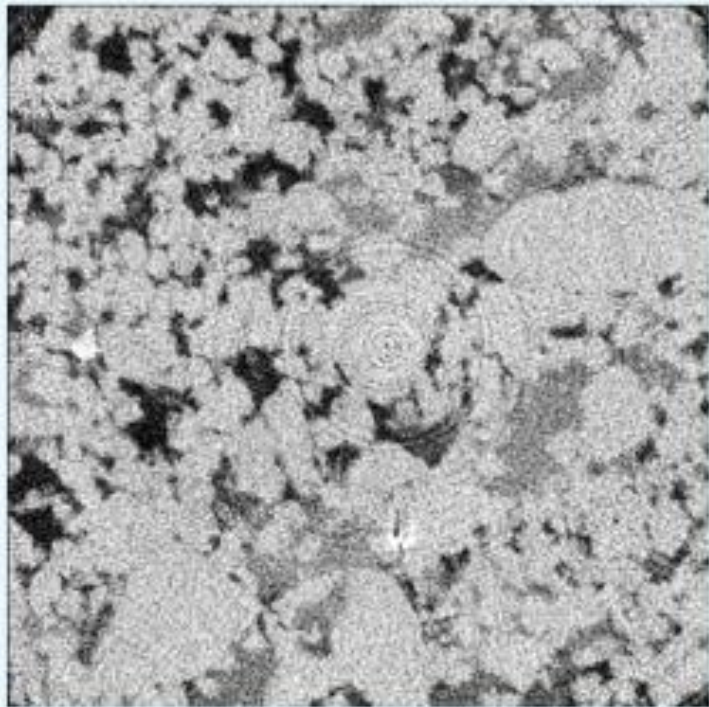
[< Previous](#)

Sample 1 of 6

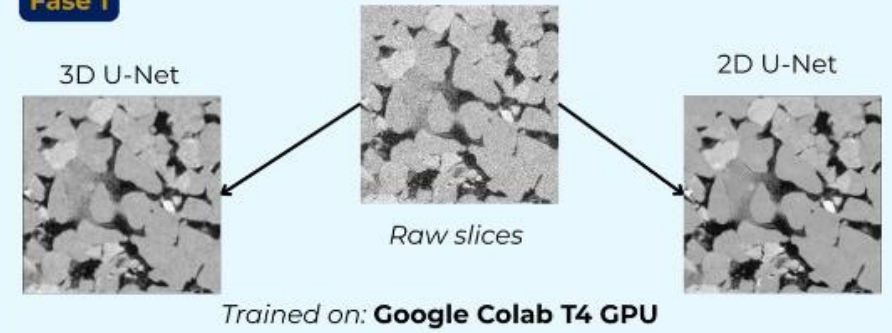
[Next >](#)

Machine Learning for Image Processing

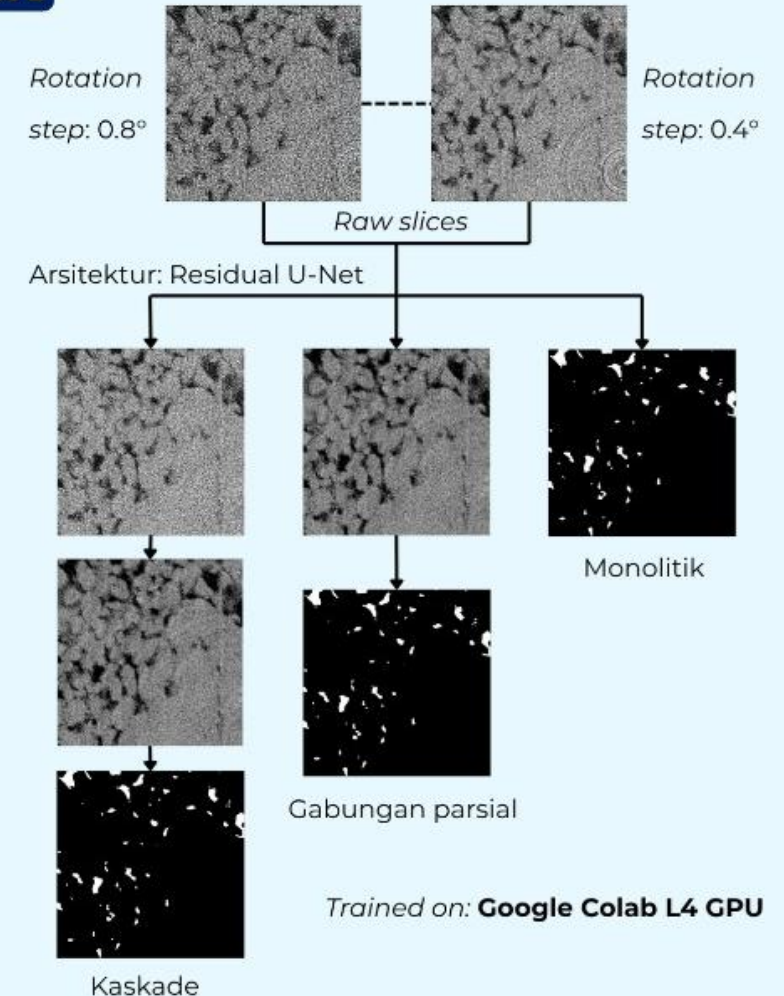
Digital Image Quality Enhancement of Rock Pore Microstructure Using ML Based Method (U-Net)



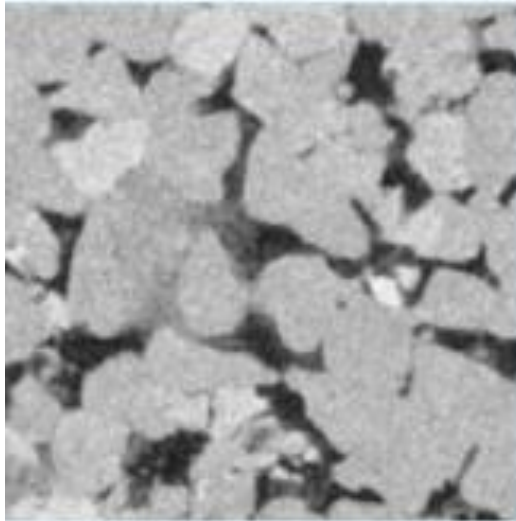
Fase 1



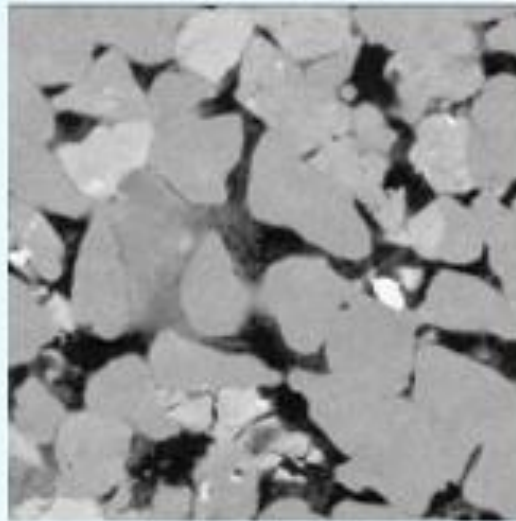
Fase 2



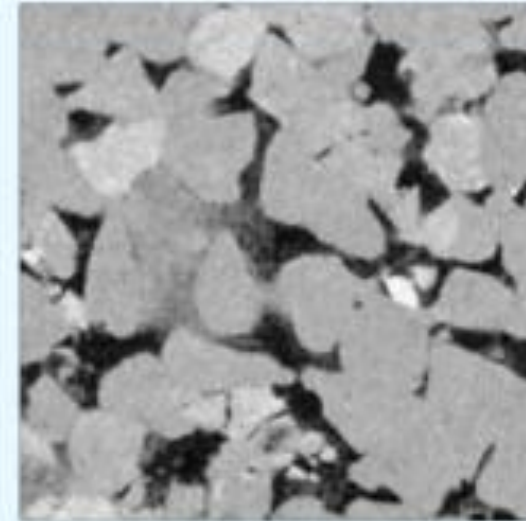
Gaussian blur



2D U-Net



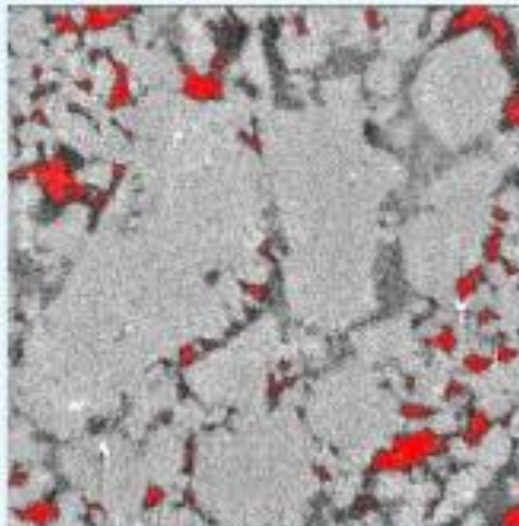
3D U-Net



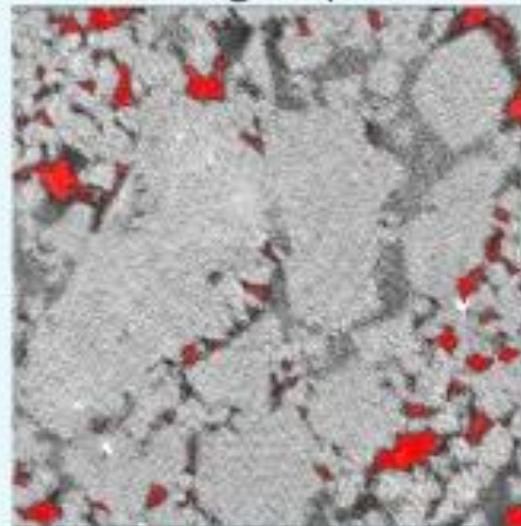
2D U-Net mencapai **error estimasi porositas terendah (3,70%)** dengan waktu pelatihan tersingkat (46 menit), unggul dibandingkan 3D U-Net.

Implementasi ketiga skema pada sampel batupasir L 1024³

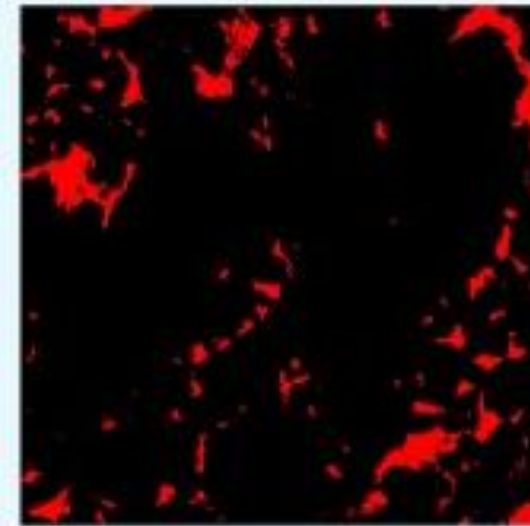
Kaskade



Gabungan parsial

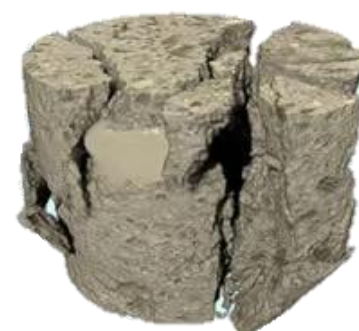


Monolitik



Instrument Development

In-Situ XRM Uniaxial Compressive Test





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